Xingyu Su

Ph.D. Student | Department of Computer Science & Engineering | Texas A&M University

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Research Interest

My research focuses on **Reinforcement Learning (RL)**-driven alignment and inference-time scaling for generative models. I develop frameworks to scale model performance through **dynamic search** and **iterative distillation** in high-dimensional, complex-reward spaces (e.g., ICLR, ICML). My expertise lies in optimizing both test-time compute and post-training efficiency to push the boundaries of model reasoning and controllable generation.

Education

Texas A&M University — College Station, TX
Ph.D. in Computer Science & Engineering — *Sep 2023 – Aug 2027 (Expected)*
- **Advisor:** Prof. Shuiwang Ji

Shanghai Jiao Tong University — Shanghai, China
Master of Science in Electronic and Information Engineering — *Sep 2020 – Feb 2023*
- **Advisor:** Prof. Yanyan Shen

Shanghai Jiao Tong University — Shanghai, China
Bachelor of Science in Mechanical Engineering — *Sep 2016 – Aug 2020*

Publications

- **Iterative Distillation for Reward-Guided Fine-Tuning of Diffusion Models in Biomolecular Design.**
  International Conference on Learning Representations (ICLR 2026).
  Xingyu Su*, Xiner Li*, Masatoshi Uehara*, Sunwoo Kim, Yulai Zhao, Gabriele Scalia, Ehsan Hajiramezanali, Tommaso Biancalani, Degui Zhi, Shuiwang Ji.
- **A Joint Diffusion Model with Pre-Trained Priors for RNA Sequence–Structure Co-Design.**
  International Conference on Learning Representations (ICLR 2026).
  Xiner Li, Masatoshi Uehara, **Xingyu Su**, Gabriele Scalia, Shuiwang Ji.
- **Learning to Discover Regulatory Elements for Gene Expression Prediction.**
  International Conference on Learning Representations (ICLR 2025, Oral Presentation – top 1.8% of accepted papers).
  Xingyu Su*, Haiyang Yu*, Degui Zhi, and Shuiwang Ji.
- **Reward-Guided Iterative Refinement in Diffusion Models at Test-Time with Applications to Protein and DNA Design.**
  International Conference on Machine Learning (ICML 2025).
  Masatoshi Uehara*, **Xingyu Su***, Yulai Zhao, Xiner Li, Aviv Regev, Shuiwang Ji, Sergey Levine, Tommaso Biancalani.
- **Dynamic Search for Inference-Time Alignment in Diffusion Models.**
  arXiv preprint arXiv:2503.02039 (2025).
  Xiner Li*, Masatoshi Uehara*, **Xingyu Su**, Gabriele Scalia, Tommaso Biancalani, Aviv Regev, Sergey Levine, Shuiwang Ji.
- **Language Models for Controllable DNA Sequence Design.**
  Transactions on Machine Learning Research (TMLR 2025).
  Xingyu Su, Xiner Li, Yuchao Lin, Ziqian Xie, Degui Zhi, Shuiwang Ji.
- **Autonomous Agents for Scientific Discovery: Orchestrating Scientists, Language, Code, and Physics.**
  arXiv preprint arXiv:2510.09901 (2025).
  Lianhao Zhou, Hongyi Ling, Cong Fu, Yepeng Huang, Michael Sun, Wendi Yu, Xiaoxuan Wang, Xiner Li, **Xingyu Su**, Junkai Zhang, Xiusi Chen, Chenxing Liang, Xiaofeng Qian, Heng Ji, Wei Wang, Marinka Zitnik, Shuiwang Ji.

- **Generative Artificial Intelligence for Biology: Toward Unifying Models, Algorithms, and Modalities.**
  ChemRxiv preprint doi:10.26434/chemrxiv.10002066/v2 (2026).
  Xiner Li*, **Xingyu Su***, Yuchao Lin, Chenyu Wang, et al., Wei Wang, Shuiwang Ji.

Professional Experience

The DIVE Lab, Texas A&M University — Sep 2023 – Present
Graduate Student Researcher — *College Station, TX*

Meituan — Feb 2023 – Jul 2023
Recommendation Algorithm Engineer (Full-time) — *Shanghai, China*

Intel — Aug 2022 – Dec 2022
Machine Learning Engineer (Intern) — *Shanghai, China*

Shanghai Jiao Tong University — 2019
Teaching Assistant - VM382 Mechanical Behavior of Materials — *Shanghai, China*

Awards & Honors

Graduate Research Excellence, TAMU CSE, one per year	2025
Undergraduate Merit Scholarship	2019
Fuda Scholarship	2019

Services | Reviewers

International Conference on Machine Learning (ICML)	2026
International Conference on Learning Representations (ICLR)	2026
Annual Conference on Neural Information Processing Systems (NeurIPS)	2025
Conference on Information and Knowledge Management (CIKM)	2025,2024